



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

EXPERIMENT :1

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Semester: V

Subject Name: ADBMS

Subject Code: 23CSP-333

1.Aim

Easy-Level Problem

Problem Title: Author-Book Relationship Using Joins and Basic SQL Operations Procedure (Step-by-Step):

- Design two tables — one for storing author details and the other for book details.
- Ensure a foreign key relationship from the book to its respective author.
- Insert at least three records in each table.
- Perform an INNER JOIN to link each book with its author using the common author ID.
- Select the book title, author name, and author's country.

Sample Output Description: When the join is performed, we get a list where each book title is shown along with its author's name and their country.

Medium-Level Problem

Problem Title: Department-Course Subquery and Access Control Procedure (Step-by-Step):

- Design normalized tables for departments and the courses they offer, maintaining a foreign key relationship.
- Insert five departments and at least ten courses across those departments.
- Use a subquery to count the number of courses under each department.
- Filter and retrieve only those departments that offer more than two courses.
- Grant SELECT-only access on the courses table to a specific user.

Sample Output Description: The result shows the names of departments which are associated with more than two courses in the system.

2. Tools Used: SQL Server Management Studio

3. Query

Easy Level Problem

```
create table tbl_author(author_id integer primary key , author_name varchar(50),country varchar(50)
);
```

```
insert into tbl_author(author_id, author_name, country) values(1011, 'Abdul Kalam', 'India');
```

```
insert into tbl_author(author_id, author_name, country) values(1022, 'J.K. Rowling', 'UK');
```

```
insert into tbl_author(author_id, author_name, country) values(1033, 'Chetan Bhagat', 'India');
```

```
select * from tbl_author;
```

```
create table tbl_book(book_id integer primary key, book_name varchar(50), author_id
INTEGER,foreign key(author_id) references tbl_author(author_id));
```

```
insert into tbl_book(book_id, book_name,author_id) values (11, 'Harry Potter',1022);
```

```
insert into tbl_book(book_id, book_name,author_id) values (12, 'Panchtantra', 1033);
```

```
insert into tbl_book(book_id, book_name, author_id) values (13, 'Wings of Fire',1011);
```

```
select * from tbl_book;
```

```
select b.book_name as [book_title], a.author_name, a.country
```

```
from tbl_book as b
```

```
inner join
```

```
tbl_author as a
```

```
on
```

```
b.author_id = a.author_id;
```

Medium Level Problem

```
create table tbl_dept (dept_id integer PRIMARY KEY,dept_name varchar(50));
```

```
insert into tbl_dept (dept_id, dept_name) values(101, 'CSE');
```

```
insert into tbl_dept (dept_id, dept_name) values(102, 'ECE');
```

```
insert into tbl_dept (dept_id, dept_name) values(103, 'Civil');
```

```
insert into tbl_dept (dept_id, dept_name) values(104, 'Mechanical');
```

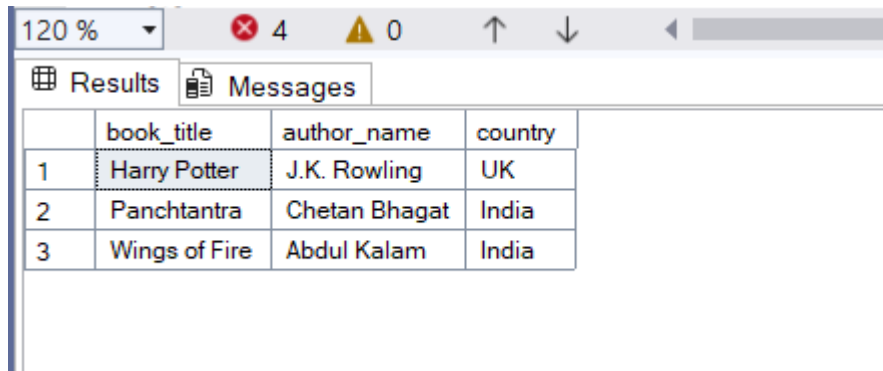
```
insert into tbl_dept (dept_id, dept_name) values(105, 'IT');
```

```
create table tbl_course (course_id integer PRIMARY KEY,course_name varchar(50),dept_id integer,  
                        FOREIGN KEY (dept_id) REFERENCES tbl_dept(dept_id));  
  
insert into tbl_course (course_id, course_name, dept_id) values(1, 'Java', 101);  
insert into tbl_course (course_id, course_name, dept_id) values(2, 'C++', 101);  
insert into tbl_course (course_id, course_name, dept_id) values(3, 'Physics', 102);  
insert into tbl_course (course_id, course_name, dept_id) values(4, 'DBMS', 105);  
insert into tbl_course (course_id, course_name, dept_id) values(5, 'DSA', 101);  
insert into tbl_course (course_id, course_name, dept_id) values(6, 'Maths', 102);  
insert into tbl_course (course_id, course_name, dept_id) values(7, 'CAD', 103);  
insert into tbl_course (course_id, course_name, dept_id) values(8, 'Thermodynamics', 104);  
insert into tbl_course (course_id, course_name, dept_id) values(9, 'Web Dev', 105);  
insert into tbl_course (course_id, course_name, dept_id) values(10, 'Python', 105);
```

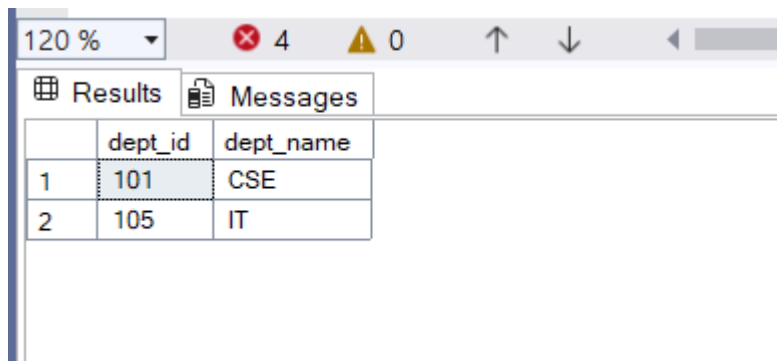
```
select * from tbl_dept;  
select * from tbl_course;
```

```
select dept_id,dept_name  
from tbl_dept  
where dept_id IN (  
    select dept_id  
    from tbl_course  
    group by dept_id  
    having count(*) > 2);  
grant select on tbl_course to username;
```

4. Output:



	book_title	author_name	country
1	Harry Potter	J.K. Rowling	UK
2	Panchtantra	Chetan Bhagat	India
3	Wings of Fire	Abdul Kalam	India



	dept_id	dept_name
1	101	CSE
2	105	IT

5. Learning Outcomes

- Acquire practical skills in setting up a primary key to uniquely identify records.
- Learn to create and enforce foreign key relationships to maintain data integrity between related tables (e.g., Books → Authors).
- Develop the ability to use INNER JOIN to combine data from multiple tables based on a shared key (e.g., author_id).
- Understand how to design normalized relational tables with foreign key constraints for realworld entities like departments and courses.